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MILITARY SPECIFICATION

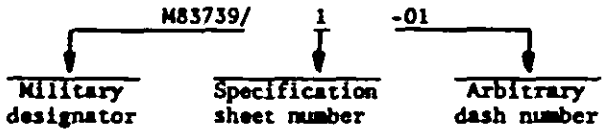
SWITCHES, ANTENNA, RADIO FREQUENCY, SOLID-STATE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for solid-state antenna switches for use at radio frequencies (see 6.1).

1.2 Military part number. The military part number shall be as follows:



2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallet, Materials-handling, Wood (General Construction Requirement).
QQ-S-781	-	Strapping, Steel, Flat and Seals.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape: Packaging, Waterproof.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

MILITARY

MIL-P-116	-	Preservation-Packaging, Methods of.
MIL-C-3643	-	Connectors, Coaxial, Radio Frequency, Series HN, and Associated Fittings, General Specification For.
MIL-C-5541	-	Chemical Conversion Coating on Aluminum and Aluminum Alloys.
MIL-P-14072	-	Finishes for Ground Signal Equipment.
MIL-S-19500	-	Semiconductor Device, General Specification For..
MIL-M-38510	-	Microcircuits, General Specification For.
MIL-C-39012	-	Connectors, Coaxial, Radiofrequency; General Specification For.

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SPECIFICATIONS - Continued

MILITARY - Continued

MIL-C-45662	-	Calibration System Requirements.
MIL-P-55110	-	Printed Wiring Boards.
MIL-C-83723	-	Connectors, Electric, Circular, Environment Resisting, General Specification For.
MIL-S-83739/1	-	Switch, Antenna, Radio Frequency, Solid-State, SA-1474A/A.

STANDARDS

FEDERAL

Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads for 40" x 48" Pallets.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	-	Reliability Tests, Exponential Distribution.
MIL-STD-785	-	Reliability Program for Systems and Equipment, Development and Production.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Qualification. Antenna switches furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.6 and 6.3).

3.3 Reliability. Reliability of switches furnished under this specification shall be established and maintained in accordance with the procedures and requirements specified in MIL-STD-781 and MIL-STD-785.

3.4 Materials. The materials shall be as specified herein. However, when materials are other than specified herein, a material shall be used which will enable the switches to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product (see 4.4).

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3.4.1 Finish. If the body of a switch is painted, it shall be painted with a semi-gloss or dull black enamel finish in accordance with type II of MIL-F-14072. Unless otherwise specified (see 6.2), all aluminum not intended to make electrical contact shall be treated in accordance with MIL-C-5541, class 3.

3.4.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Fungus. Materials used in the construction of switches shall be fungus inert in accordance with requirement 4 of MIL-STD-454.

3.5 Design and construction.

3.5.1 Dimensions, mounting, and weight. Envelope dimensions, mounting, mounting dimensions, connector arrangement, and weight shall be as specified (see 3.1).

3.5.2 Micro-electronic circuits. Micro-electronic circuits shall be in accordance with MIL-M-38510, class B, level A, minimum. The design shall incorporate a minimum number of components.

3.5.3 Semiconductors. Semiconductors shall be in accordance with MIL-S-19500, TX or TXV level.

3.5.4 Printed circuit boards. Printed circuit boards shall be in accordance with MIL-P-55110.

3.5.5 Connectors.

3.5.5.1 Radio frequency (RF) connectors. Connectors for a specific switch shall be as required (see 3.1). Material, design, and construction of the types listed below shall conform to the specifications as follows:

Type of RF connector	Specification
N	MIL-C-39012
TNC	MIL-C-39012
BNC	MIL-C-39012
SMA	MIL-C-39012
SMC	MIL-C-39012
SC	MIL-C-39012
C	MIL-C-39012
HN	MIL-C-3643

3.5.5.2 Power connectors. Connectors for a specific switch shall be as required (see 3.1). Materials, design, and construction shall conform to MIL-C-83723.

3.5.6 Configuration. Switches shall be furnished with the number of positions specified (see 3.1).

3.5.7 Sequence. Unless otherwise specified (see 3.1), switches shall have a break-before-make or make-before-break sequence of operation.

3.5.8 Nominal impedance. Nominal impedance shall be as specified (see 3.1).

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3.5.9 Maintainability. Unless otherwise specified (see 3.1), switches furnished under this specification shall be the nonrepairable types. Mounting provisions shall insure that a minimal amount of time and effort is required for removal and replacement, using standard tools only.

3.5.10 Housings. Switch housings shall be as identified below and as specified (see 3.1).

(a) Enclosed (to exclude humidity) - type E.
(b) Hermetically sealed - type H.
(c) Immersion-proof - type I.

3.5.11 Frequency range. Switches shall be designed to operate throughout the specified frequency range (see 3.1).

3.5.12 Electromagnetic interference. Interference control requirements for switches shall be in accordance with MIL-STD-461.

3.6 Performance.

3.6.1 Switching rate. When switches are tested as specified in 4.8.2, the switching rate shall not exceed the value specified (see 3.1).

3.6.2 Primary input power. When switches are tested as specified in 4.8.2, the primary input power shall not exceed the value specified (see 3.1).

3.6.3 Transition (overlap) time. When switches are tested as specified in 4.8.3, the transition time shall not exceed the value specified (see 3.1). The transition time is the period when the outputs are considered in parallel and the isolation is minimum.

3.6.4 Voltage standing wave ratio (VSWR). When switches are tested as specified in 4.8.4, the VSWR shall not exceed the value(s) specified (see 3.1).

3.6.5 Isolation. When switches are tested as specified in 4.8.5, the isolation between outputs shall be not less than the value specified (see 3.1).

3.6.6 Insertion loss. When switches are tested as specified in 4.8.6, the insertion loss shall not exceed the value specified (see 3.1).

3.6.7 RF power handling capability. When switches are tested as specified in 4.8.7, with rated peak power, and rated average power (when specified, see 3.1) passing through the switch, no evidence of breakdown shall be evident. Following each test, the VSWR value (see 3.1) shall be no greater than specified (see 3.1). Unless otherwise specified (see 3.1), switches shall be required to switch under power.

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3.6.8 Temperature shock. When switches are tested as specified in 4.8.8, the VSWR, isolation, insertion loss, primary input power, switching rate, and transition time shall be as specified (see 3.1).

3.6.9 Vibration. When switches are tested as specified in 4.8.9, the VSWR, isolation insertion loss, primary input power, switching rate, and transition time shall be as specified (see 3.1).

3.6.10 Humidity (applicable to type E only). When switches are tested as specified in 4.8.10, the VSWR, isolation, insertion loss, primary input power, switching rate, and transition time shall be as specified (see 3.1).

3.6.11 Hermetic seal (applicable to type H only). When switches are tested as specified in 4.8.11, the leakage rate shall not exceed the applicable value specified below. The volume shall be computed using the external dimensions of the switch housing, disregarding any mounting (screw, stud, etc.,).

Sealed volume of case	Maximum allowable leakage
Greater than 2 cubic inches	1×10^{-6} atm cc/sec
2 cubic inches or less	1×10^{-8} atm cc/sec

3.6.12 Immersion (applicable to type I only). When switches are tested as specified in 4.8.12, the VSWR, isolation, insertion loss, primary input power, switching rate, and transition time shall be as specified (see 3.1).

3.6.13 Salt fog. When switches are tested as specified in 4.8.13, the VSWR, isolation, insertion loss, primary input power, switching rate, and transition time shall be as specified (see 3.1).

3.6.14 Shock. When switches are tested as specified in 4.8.14, the VSWR, isolation, insertion loss, primary input power, switching rate, and transition time shall be as specified (see 3.1).

3.6.15 Temperature-altitude. When switches are tested as specified in 4.8.15, the VSWR, isolation, insertion loss, primary input power, switching rate, and transition time shall be as specified (see 3.1).

3.6.16 Acceptance reliability. When tested as specified in 4.8.16, unless otherwise specified (see 3.1), switches shall provide a minimum acceptable mean-time-between failure (MTBF) of 10,000 hours.

3.7 Marking (see 3.1). Switches shall be marked with the manufacturer's code identification and the military part number (see 3.1). The date of manufacturer of the item shall be marked on the part. The date code shall be as specified in MIL-STD-1285.

(a) Switches shall be marked with the type of supply, the operating voltage, and the frequency.
(b) Input and output connections shall be identified in a logical manner.

The marking shall remain legible after completing all environmental tests.

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3.8 Workmanship. Switches shall be processed in such a manner as to be uniform in quality and shall be free from corrosion and other defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-785. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.3 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.4).
(b) Qualification inspection (see 4.6).
(c) Quality conformance inspection (see 4.7).

4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the switches, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE I. Materials inspection.

Material	Requirement paragraph	Applicable specification
Finishes	3.4.1	MIL-C-5541, MIL-F-14072
Dissimilar metals	3.4.2	MIL-STD-889
Fungus	3.4.3	MIL-STD-454
Micro-electronic circuits	3.5.2	MIL-M-38510
Semiconductors	3.5.3	MIL-S-19500
Printed circuit boards	3.5.4	MIL-P-55110
Connectors	3.5.5	MIL-C-39012, MIL-C-3643, MIL-C-83723

4.5 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

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4.6 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.6.1 Sample size. Four switches shall be subjected to qualification inspection. Each switch with a different basic design shall be qualified separately.

4.6.2 Inspection routine. The sample shall be subjected to the inspections specified in table II, in the order shown. All sample units shall be subjected to the inspections of group I. The sample shall then be divided into two groups of two units each. The sample units shall then be subjected only to the inspections indicated for the particular group.

TABLE II. Qualification inspection.

Examination or test	Requirement paragraph	Test paragraph
Group I (all samples)		
Visual and mechanical examination	3.4, 3.5.1, and 3.8	4.8.1
Marking	3.7	4.8.1
Switching rate	3.6.1	4.8.2
Primary input power	3.6.2	4.8.2
Transition time	3.6.3	4.8.3
Voltage standing wave ratio	3.6.4	4.8.4
Isolation	3.6.5	4.8.5
Insertion loss	3.6.6	4.8.6
RF power handling capability	3.6.7	4.8.7
Temperature shock	3.6.8	4.8.8
Vibration	3.6.9	4.8.9
Group II (2 samples)		
Salt fog	3.6.13	4.8.13
Shock	3.6.14	4.8.14
Temperature-altitude	3.6.15	4.8.15
Group III (2 samples)		
Humidity (applicable to type E only)	3.6.10	4.8.10
Hermetic seal (applicable to type H only)	3.6.11	4.8.11
Immersion (applicable to type I only)	3.6.12	4.8.12

4.6.3 Failures. No failures shall be allowed for qualification inspection; a failure shall be anything that does not meet the requirements of the specification.

4.6.4 Reliability qualification inspections. Seven switches shall be tested for reliability in accordance with test level G of MIL-STD-781. Two of the seven switches shall continue through a longevity test in accordance with MIL-STD-781. Longevity test shall not be a criteria for qualification of the switch.

4.6.4.1 Accept-reject criteria. Acceptance shall be judged on the basis of the table entitled Accept- Reject Criteria For: Test Plan III of MIL-STD-781.

4.6.4.2 Retention of qualification. To retain qualification, the supplier shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery, (group A), indicating as a minimum the number of

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lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

(b) A summary of the results of tests performed for qualification verification inspection, (group B), including the number and mode of failures. The summary shall include results of all qualification verification inspection tests performed on completed lots during the 12-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time that the inspection data indicates noncompliance of the product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of each switch to testing in accordance with the qualification inspection requirements.

4.7 Quality conformance inspection.

4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A.

4.7.1.1 Inspection lot. An inspection lot shall consist of all switches of the same part number, produced under essentially the same conditions, and offered for inspection at one time.

4.7.1.2 Group A inspection. Group A inspection shall consist of the examination and tests specified in table III, in the order shown.

TABLE III. Group A inspection.

Examination or test	Requirement paragraph	Test paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Subgroup I				
Visual and mechanical examination	3.4, 3.5.1, and 3.8	4.8.1	.65	1.5
Marking	3.7	4.8.1	.65	1.5
Switching rate	3.6.1	4.8.2	.65	---
Primary input power	3.6.2	4.8.2	.65	---
Transition time	3.6.3	4.8.3	.65	---
Voltage standing wave ratio	3.6.4	4.8.4	.65	---
Isolation	3.6.5	4.8.5	.65	---
Insertion loss	3.6.6	4.8.6	.65	---
Subgroup II				
Acceptance reliability	3.6.16	4.8.16	---	---

4.7.1.2.1 Sampling plan. For subgroup I, statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level G-2. The acceptable quality level (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105. For subgroup II, the monthly sample size shall be tested. Accept-reject criteria shall be in accordance with Test Plan III of MIL-STD-781.

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4.7.1.3 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.7.2 Qualification verification inspection. Qualification verification inspection shall consist of group B. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.7.2.1.4), delivery of products which have passed group A shall not be delayed pending the results of these qualification verification inspections.

4.7.2.1 Group B inspection. Group B inspection shall consist of the tests specified in table IV, in the order shown. Group B inspection shall be made on sample units which have passed group A inspection.

TABLE IV. Group B inspection.

Test	Requirement paragraph	Test paragraph	Number of sample units to be inspected	Number of failures allowed 1/
RF power handling capability	3.6.7	4.8.7	6	1
Temperature shock	3.6.8	4.8.8		
Vibration	3.6.9	4.8.9		
Salt fog	3.6.13	4.8.13		
Shock	3.6.14	4.8.14		
Temperature-altitude	3.6.15	4.8.15		
Humidity (applicable to type E only)	3.6.10	4.8.10		
Hermetic seal (applicable to type H only)	3.6.11	4.8.11		
Immersion (applicable to type I only)	3.6.12	4.8.12		

1/ A sample unit having one or more defects shall be considered as a single failure.

4.7.2.1.1 Sampling plan. Sample units shall be selected every 6 months. Upon passing this inspection, the supplier may select sample units every 12 months. If the second level of sampling is passed two successive times, the supplier may select sample units every 24 months. In the event of a failure, sampling shall revert to the 6-month interval.

4.7.2.1.2 Failures. If the number of failures exceeds the number allowed in table IV, the sample shall be considered to have failed.

4.7.2.1.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.7.2.1.4 Noncompliance. If a sample fails to pass group B inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the Government). Group A inspection may be reinstated; however, final acceptance shall be withheld until group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

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4.7.3 Inspection of preparation for delivery. Except for commercial packaging, the sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract or purchase order (see 6.2).

4.8 Methods of examination and test.

4.8.1 Visual and mechanical examination (see 3.4, 3.5.1, 3.7, and 3.8). Switches shall be examined to verify that the materials, size, mounting, weight, design, construction, finish, marking, and workmanship are in accordance with the applicable requirements.

4.8.2 Switching rate and primary input power (see 3.6.1 and 3.6.2). The switching rate and primary input power of the switch shall be determined at the center of the specified (see 3.1) RF frequency range and the primary input voltage and frequency (when applicable) at the specified (see 3.1) nominal, lowest, and highest values.

4.8.3 Transition time (see 3.6.3). The transition time shall be measured by determining the length of time that the isolation between outputs is at the lowest value. Test shall be made with the switch energized with nominal input power and an RF signal into one RF output and a pad, detector, and oscilloscope connected to another RF output. The RF signal shall be at the center frequency of the specified frequency range.

4.8.4 Voltage standing wave ratio (see 3.6.4). The VSWR shall be determined for each switch position and when specified (see 3.1), during the period when the outputs are in parallel. Unless otherwise specified (see 3.1), swept frequency technique shall be used to determine the VSWR over the specified (see 3.1) frequency range.

4.8.5 Isolation (see 3.6.5). With the switch energized with nominal input power, the isolation between outputs shall be determined using the swept frequency technique over the specified frequency range.

4.8.6 Insertion loss (see 3.6.6). The insertion loss shall be measured with the switch stopped on each RF output. Swept frequency technique shall be used to determine the insertion loss over the specified frequency range (see 3.1).

4.8.7 RF power handling capability (see 3.6.7). With the switch energized with nominal input power, the switch shall be subjected to the specified (see 3.1) RF power and frequency for the specified period of time. During the test the switch shall be terminated with the specified load. After the test, the VSWR shall be measured as specified in 4.8.4.

4.8.8 Temperature shock (see 3.6.8). Switches shall be placed within a test chamber wherein the highest specified temperature (see 3.1) is maintained. Switches shall be exposed to this temperature for a period of 4 hours, unless otherwise specified; at the conclusion of this time and within 5 minutes, the switches shall be transferred to a chamber having the lowest specified temperature (see 3.1) for a period of 4 hours, unless otherwise specified. This constitutes one cycle of temperature shock. Switches shall be nonoperating for the entire exposure. Three complete cycles shall be performed on a continuing basis. However, either the high or the low temperature soak periods can be extended to an overnight period to prevent interruption of the transfer sequence. After the test, the switches shall be subjected to the individual tests of group I in table III.

4.8.9 Vibration (see 3.6.9). Switches shall be tested in accordance with method 514.2, category b.1, Procedure 1, table 514-2-II, figure 514.2-2, curve AR to 33 Hz and switch to curve H of MIL-STD-810 for remainder of vibration. After the test, switching rate, primary input power, transition time, VSWR, isolation, and insertion loss shall be measured as specified in 4.8.2 through 4.8.6, respectively.

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4.8.10 Humidity (applicable to type E only) (see 3.6.10). Switches shall be tested in accordance with method 507.1, procedure 1 of MIL-STD-810. After the test, switching rate, primary input power, transition time, VSWR, isolation, and insertion loss shall be measured as specified in 4.8.2 through 4.8.6, respectively.

4.8.11 Hermetic seal (applicable to type H only) (see 3.6.11). Hermetically-sealed switches shall be tested in accordance with method 112 of MIL-STD-202. The following details shall apply:

(a) Test condition letter - C (protective connector caps shall be used during test).
(b) Procedure III.
(c) Degree of leakage rate sensitivity - See 3.6.11.

4.8.12 Immersion (applicable to type I only) (see 3.6.12). Immersion-proof switches shall be tested in accordance with method 104 of MIL-STD-202. The following details shall apply:

(a) Test condition letter - A (protective connector caps shall be used during test).
(b) During test cycle and while submerged, the sample shall be subjected to a low barometric pressure of 3.44 inHg (50,000 feet altitude).
(c) Measurements - No leakage as evidenced by a continuous stream of bubbles. Bubbles which are the result of entrapped air on the exterior surface of the switch shall not be considered as an indication of leakage.

After the test, switching rate, primary input power, transition time, VSWR, isolation, and insertion loss shall be measured as specified in 4.8.2 through 4.8.6, respectively.

4.8.13 Salt fog (see 3.6.13). Switches shall be tested in accordance with method 509.1, procedure I of MIL-STD-810. After the test, switching rate, primary input power, transition time, VSWR, isolation, and insertion loss shall be measured as specified in 4.8.2 through 4.8.6, respectively.

4.8.14 Shock (see 3.6.14). Switches shall be tested in accordance with method 516.2 "procedure 1" of MIL-STD-810, figure 516.2-1, time duration 11 ms, 20 g's. After the test, switching rate, primary input power, transition time, VSWR, isolation, and insertion loss shall be measured as specified in 4.8.2 through 4.8.6, respectively.

4.8.15 Temperature-altitude (see 3.6.15). Switches shall be tested in accordance with method 504.1, procedure I of MIL-STD-810, for equipment category type 7. After the test, switching rate, primary input power, transition time, VSWR, isolation, and insertion loss shall be measured as specified in 4.8.2 through 4.8.6, respectively.

4.8.16 Acceptance reliability (see 3.6.16). Switches shall be tested in accordance with test level G of MIL-STD-781 with the following additions and exceptions:

(a) On-off cycling shall be timed to stabilize at low temperature (with no electrical operation) followed by time to stabilize at the high temperature plus 4 hours (with continuous operation at rated input voltage).
(b) The electrical performance shall be checked at interval of 100 hours by repeating the switching rate, primary input power, transition time, VSWR, isolation, and insertion loss tests. Daily tests shall be conducted to insure proper switch operation.
(c) Vibration is not required.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, or as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Cleaning. Switches shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Switches shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified (see 6.2), each switch shall be individually unit packaged in accordance with MIL-P-116, submethod IA-15 insuring compliance with the applicable requirements of that specification. The container shall conform to PPP-B-566, PPP-B-676, or PPP-B-636.

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Switches shall be clean, dry, and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, or as specified (see 6.2).

5.2.1 Level A. The packaged switches shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged switches shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged switches shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Switches, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Switches, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Switches, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size, and kind commonly used for this purpose. These unitized loads shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

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5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), each unit package, exterior container, and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.7.3.

5.4.3 Army procurements.

5.4.3.1 Level A (maximum military protection) unit packaging. MIL-P-116 submethod IC-2 shall be used in lieu of submethod IA-15.

5.4.3.2 Level A (maximum military protection) and level B (minimum military protection) packing. For level A packing the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish A. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5.2.1 and 5.2.2).

5.4.3.3 Level A and B unitization. For level A and B unitization, the fiberboard caps shall be weather resistant and softwood pallets conforming to NN-P-71, type VI, size 2 shall be used. The loads shall be bonded to the pallets by strapping conforming to QQ-S-781, type I, finish A or shrink film (see 5.2.4.1 and 5.2.4.2).

5.4.3.4 Commercial packaging. Commercial packaging (including unit packaging, packing and marking) shall be in accordance with Fed. Std. No. 356.

6. NOTES

6.1 Intended use. Switches covered by this specification are designed for use with RF coaxial transmission line systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of applicable specification sheet, and the complete part number.
(c) Specify when aluminum should not be anodized (see 3.4.1).
(d) Inspection of commercial packaging (see 4.7.3).
(e) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(f) Method of preservation, if other than submethod IA-15 (see 5.1.1.4 and 5.4.3.1).
(g) Special marking, if required (see 5.3).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Interservice Interagency Support Group for FSG 59, AFLC/MMGE, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.3.1 Copies of SD-6, "Provisions Governing Qualification," may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army -
Navy -
Air Force - 11, 80
DSA - ES

User activities:
Army -
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Preparing activity:
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DOCUMENT IDENTIFIER AND TITLE

MIL-S-83739A Switch, Antenna, Radio Frequency, Solid-State FSC 5985

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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